



Physics final

1. A) $RPM = \frac{60 \text{ seconds}}{1.33 \text{ seconds/revolutions}} = 45 \text{ rpm}$

B) Angular velocity = $\frac{2\pi \text{ radians}}{1.33 \text{ seconds}} = 4.7 \frac{\text{radians}}{\text{second}}$
 B = 45 rpm, 4.7 radians/sec

2. $a_c = r\omega^2 \frac{2\pi}{1.23} \approx 4.72 \text{ rad/s}$

$\omega = 4.7 \text{ rad/s}$ $a_c = \omega^2 r = 4.72^2 r$
 $a_c = 22.3r$ $a_c = 22.3(0.05) = 1.1 \text{ m/s}^2$

3. blood centrifuge need so be increased by a factor of 100.

as $\omega = 10$ $10(10) = 100$

D: $\omega \rightarrow 10\omega$

4. Centrifuge spins at 200 rpm, $r = 12 \text{ cm}$, $m = 10 \text{ g}$, what is centripetal force?

$\omega = \frac{200 \cdot 2\pi}{60} = 20.94 \frac{\text{radians}}{\text{second}} \rightarrow a_c = r \cdot \omega^2 = 0.12 (20.94)^2 = 52.53 \frac{\text{m}}{\text{s}^2}$

$F_c = m \cdot a_c = 0.01 (52.53) \Rightarrow 0.53 \text{ N}$

B: 0.53 N

5. in lacrosse, a ball rotates 30 rad/sec at the elbow joint 1.40 m away what is the angular velocity?

$v = r \cdot \omega \rightarrow 1.40 (30.0) = 42 \text{ m/s}$

C = 42 m/s

Unit s

6. person pushes box $\vec{f} = 40\hat{i} - 30\hat{j} \text{ N}$, box moves $\vec{x} = 5\hat{i} \text{ m}$. What is work done? What is angle between force and displacement

$w = \vec{F} \cdot \vec{x} = (40\hat{i} - 30\hat{j}) \cdot (5\hat{i}) = 200 \text{ J m}$

angle = $\sqrt{40^2 + 30^2} = 50$ $\cos \theta = \frac{200}{50 \cdot 5} = 0.8 \Rightarrow \theta = \cos^{-1}(0.8) = 37^\circ$

D = 200 J m, 37°

7. $F = 100 \text{ N}$, $d = 0.1 \text{ m}$, $m = 0.15 \text{ kg}$, $v_i = 0$, launch angle = 45°

A) $w = F \cdot d \Rightarrow (100)(0.1) = 100 \text{ J}$

B) if the ball was initially stationary then work = 0 J
 $KE_I = \frac{1}{2} m v_i^2 = \frac{1}{2} (0.15 \text{ kg}) \cdot (0)^2 = 0 \text{ J}$

C) $KE = \frac{1}{2} m (v_f^2)$

$v_f = \sqrt{\frac{2KE}{m}}$, $v_f \sqrt{\frac{2 \cdot 100}{0.15}} = \sqrt{1333.3} \Rightarrow 36.5 \text{ m/s}$

D) $v_f = 36.5 \text{ m/s}$ $\theta = 45^\circ$ $g = 9.81 \text{ m/s}^2$ $\sin 90^\circ = 1$

$R = \frac{(36.5)^2 \cdot 1}{9.81} = \frac{1332.25}{9.81} \approx 136 \text{ m}$

Unit 5 question 3 | a person pushing a 40 kg box over 4 m on a smooth floor with $\mu_k = 0.05$. what is work done by friction?

$$\text{Force of friction } (F_f) = \mu_k \cdot m \cdot g \Rightarrow (0.05)(40)(9.81) = 19.6 \text{ N}$$

$$W_f = (19.6 \text{ N})(40 \text{ m}) = 784 \text{ J}$$

$$\text{Power generated at 30 seconds } P = \frac{W}{t} \quad P = \frac{784 \text{ J}}{30 \text{ seconds}} \Rightarrow 26.13 \text{ W}$$

Unit 5 ques 4 | what is the cost of operating 3.00 W clock for a year with cost = 0.09 cents per kWh

$$\text{Power of clock} = P = 3.00 \text{ W} \rightarrow 0.003 \text{ kW}$$

$$t = 1 \text{ year} = 365 \text{ days} \times \frac{24 \text{ hours}}{1 \text{ day}} = 8760 \text{ hours}$$

$$\text{Cost of electricity} = 0.09 \text{ USD/kWh}$$

$$E = P \cdot t \Rightarrow (0.003 \text{ kW}) \cdot (8760 \text{ hours}) = 26.28 \text{ kWh}$$

$$\text{Cost} = E \cdot \text{cost per kWh} \Rightarrow 26.28 \text{ kWh} (0.09) = 2.37 \text{ USD}$$

$$D = 2.33 ?$$

Unit 6 ques 1 | $m_1 = m_2 = 20 \cdot 10^{-25} \text{ kg}$, $v_1 = 350 \text{ m/s}$, $v_2 = -350 \text{ m/s}$ $v_f = ?$

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

$$(\cancel{20 \times 10^{-25} \text{ kg}})(350 \text{ m/s}) + (\cancel{20 \times 10^{-25} \text{ kg}})(-350 \text{ m/s}) = (\cancel{20 \times 10^{-25} \text{ kg}} + \cancel{20 \times 10^{-25} \text{ kg}}) \cdot v_f$$

$$v_f = 0$$

$$C = 0 \text{ m/s}^{-1}$$

Unit 6 ques 2 | given: $m_1 = m_2 = 20 \times 10^{-25} \text{ kg}$, speed = $v_1 = v_2 = 350 \text{ m/s}$ angle = 45°

$$m_{\text{total}} = m_1 + m_2 = 2(20 \times 10^{-25} \text{ kg}) = (40 \times 10^{-25} \text{ kg})$$

$$p_1 = m_1 \cdot v_1 = (20 \cdot 10^{-25})(350) = 7000 \times 10^{-25} \text{ m/s}$$

$$p_2 = m_2 \cdot v_2 = 7000 \cdot 10^{-25} \text{ kg m/s}$$

$$p_{\text{total}} = \sqrt{p_1^2 + p_2^2 + 2 \cdot p_1 \cdot p_2 \cdot \cos(45^\circ)}$$

$$p_{\text{total}} = \sqrt{(7000 \times 10^{-25} \text{ kg m/s})^2 + (7000 \times 10^{-25} \text{ kg m/s})^2 + 2 \cdot (7000 \times 10^{-25} \text{ kg m/s})^2 \cdot \cos 45^\circ}$$

$\hookrightarrow$

$$\sqrt{49 \times 10^{-46} + 49 \times 10^{-46} + 2 \cdot 49 \times 10^{-46} \cdot \frac{\sqrt{2}}{2}}$$

$$\approx 12435 \times 10^{-25}$$

$$v_f = \frac{p_{\text{total}}}{m_{\text{total}}} = \frac{12435 \times 10^{-25}}{40 \times 10^{-25}} = 310.875 \text{ m/s}$$

Unit 6 ques 3 | $F = 4000 \text{ N}$ $t = 0.200 \text{ s}$ $v_i = 2.80 \text{ m/s}$ $m = 200 \text{ kg}$

$$A = J = F \cdot t \Rightarrow J = 4000 \cdot 0.200 \text{ s} = 800 \text{ N/s}$$

$$b = J = \Delta p = m \cdot \Delta v \quad J = 800 \text{ N/s}, m = 200 \text{ kg} \quad v_i = 2.80 \text{ m/s}$$

$$\Delta v = v_f - v_i$$

$$v_f = \frac{J}{m} + v_i$$

$$v_f = \frac{800 \text{ N/s}}{200 \text{ kg}} + 2.80 \text{ m/s}$$

$$-4 \frac{\text{N}}{\text{kg}} + 2.80 \text{ m/s} = -1.20 \text{ m/s}$$

Unit 6 ques 4 | This interaction is C: Elastic

Unit 6 ques 5: $m_1 = 30,000 \text{ kg}$ $v_1 = 0.850 \text{ m/s}$ $m_2 = 110,000 \text{ kg}$ $v_2 = 0 \text{ m/s}$

A) $m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$
 $((30,000 \text{ kg})(0.850 \text{ m/s})) + ((110,000 \text{ kg})(0)) = 30,000 + 110,000 \text{ kg} \cdot v_f$
 $25,500 = 140,000 \cdot v_f$
 $v_f = \frac{25500}{140,000} = 0.182 \text{ m/s}$

B) $KE_i = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$
 $= \frac{1}{2} (30,000 \text{ kg})(0.850)^2 + \frac{1}{2} (110,000) \cdot 0^2$
 $= 15,000 \cdot 0.7225 = 10,837.5 \text{ J}$

$KE_{\text{final}} = \frac{1}{2} (m_1 + m_2) v_f^2$
 $= \frac{1}{2} (140,000 \cdot (0.182)^2$
 $= 70,000 \cdot 0.0331 = 2,318.7 \text{ J}$

Kinetic Energy lost

$\hookrightarrow KE_i - KE_f$

$\Delta KE = 10,837.5 - 2,318.7 = 8518.8 \text{ J}$

Unit 7 Question 1 | $\text{mass bricks} = 40 \text{ kg}$ $\text{distance} = 3 \text{ m}$ $\text{mass car} = 60 \text{ kg}$ $\text{distance}_{\text{car}} = 2 \text{ m}$ $g = 9.81 \text{ m/s}^2$

$T_b = 3 \cdot (40 \text{ kg})(9.81) = 3 \cdot 392 = 1176 \text{ N/m}$

$T_c = 2 \cdot (60 \text{ kg})(9.81 \text{ g}) = 2.388 = 1176 \text{ N}$

motionless A

Unit 7 question 2 | $L = 2.00 \text{ m}$, $m_{\text{plate}} = 20.0 \text{ kg}$, $W = m \cdot g = (20.0)(9.81) 196 \text{ N}$, Center of gravity 1.00 m $\text{Distance } F_1 = 0.00 \text{ m}$ $\text{Distance } F_2 = 0.50 \text{ m}$

$T_{F_2} = F_2 \cdot X_2 = F_2 \cdot 0.50$

$T_W = 196 \text{ N} \cdot 1.00 \text{ m} = 196 \text{ N/m}$

$F_2 = \frac{196 \text{ N}}{0.50} = 392 \text{ N}$

$F_2 = F_1 + W \rightarrow 392 \text{ N} = F_1 + 196 \rightarrow F_1 = 392 - 196 = 196 \text{ N}$

Force $F_1 = 196 \text{ N}$ pushing down one end

Force $F_2 = 392 \text{ N}$ holding up 0.5 m

Unit 8 ques 1 $\text{mass}_{\text{gear}} = 1 \text{ kg}$ $r = 8 \text{ cm}$ (0.03 m) $\alpha = 2 \text{ rad/s}^2$

A) $I = \frac{1}{2} m r^2$

$\hookrightarrow I = \frac{1}{2} \cdot (1 \text{ kg})(0.08 \text{ m})^2$

moment of inertia $= 0.0032 \text{ kg/m}^2$

B) $T = I \cdot a \rightarrow 0.0032(2)$

$\tau = 0.0064 \text{ N/m}$

Unit 8 ques 2 | Position vector of the pedal: $\vec{r} = -2\hat{i} + 2\hat{j} \text{ cm} = -0.20\hat{i} + 0.20\hat{j} \text{ m}$ Force of pedal: $\vec{F} = 12\hat{i} \text{ N}$

A) $\vec{r} = \vec{r}(\vec{r})$
 $\vec{r} = (-0.20, 0.20, 0) \text{ m}$
 $\vec{F} = (12, 0, 0) \text{ N}$